

DH2323 Project Specification, Version 2

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Status: As-shipped specification, written after implementation, evaluation, and report completion

Why a v2 exists

Version 1 (April 22, 2026) was a pre-development draft submitted for early professor feedback, before any code existed. Several of its plans changed during development, sometimes at the professor's direction. This version describes what was actually built, not what was originally proposed. The original is preserved unchanged for reference as project-spec-v1-original-2026-04-22.pdf.

Project Title

Let Me Ask Charlie: A Procedural Real-Time Lighting and Shadow Pipeline for Browser-Based Interactive Horror.

Working title during early development was Charlie / Chat. The shipped title matches the report and the published game.

Concept

Let Me Ask Charlie is a short interactive horror vignette built around a custom real-time lighting and shadow pipeline running in the browser via p5.js and GLSL fragment shaders. The renderer itself produces the horror: the antagonist, Charlie Chat, has no sprite of its own and is rendered entirely as a signed distance field whose soft shadow grows and spreads as it approaches the player.

The player is pursued through a single room, not the three originally planned (see Changes from v1 below). The antagonist is a dark, glowing-eyed silhouette rather than a white one, since its visual presence comes from the SDF shadow system rather than an authored sprite. The story is based on an original narrative, Let Me Ask Charlie, reframed into a horror context. A single binary player choice (YES / NO) creates two branches that converge at the same ending, since the narrative branch itself was confirmed by Professor Christopher Peters not to be part of the graded technical contribution.

Technical Direction

Core technical contribution: a fully procedural real-time lighting and shadow pipeline, with no pre-authored normal maps, sprite sheets, or image assets anywhere in the project.

- Procedural environment normal mapping, not sprite normal mapping: surface normals are derived per-fragment from a greyscale height map using a Sobel gradient kernel, driving Lambertian (N·L) diffuse shading against a moving point light.
- Shared SDF soft-shadow raymarching, covering both the moving antagonist and static furniture, built on Quilez's soft-shadow near-miss estimator and Hart's sphere-tracing principle.
- All characters are flat, procedural 2D primitives, with no normal mapping on characters at all. This was a deliberate scope decision made with Professor Christopher Peters, trading character visual fidelity for full traceability of every visual element to code.
- Three-stage chase pacing: a slow opening giving the player room to explore before pressure begins, a ramp toward a 20 to 30 second target duration, and a late rubber-band speed spike if the player is evading well as time runs low. Tuned after playtesting an earlier flat-ramp version that felt wrong.
- A four-effect post-processing pass: bloom, Reinhard tone mapping, a shadow-tint colour grade, and hash-based film grain.

- Spatial procedural audio: a drone synthesized directly from raw Web Audio API oscillator, gain, and stereo-panner nodes, panned to the antagonist's position and intensifying with proximity.

Changes from v1

- One room, not three. Following feedback from Professor Christopher Peters at the help session on April 24: depth on the rendering technique was prioritized over breadth of levels.
- Environment gets normal mapping; characters do not. The reverse of the original plan. This was also a direct scope decision from Professor Christopher Peters, and reduced art production time substantially while keeping every visual element procedural.
- Chromatic aberration was planned but never built. Disclosed as a known, cut scope item in the final report rather than silently dropped.
- Three-stage chase pacing is new. Not part of the original spec at all, added after the first working version proved either unwinnable or, once fixed, paced wrong.

Visual Reference

Ernst Haas: desaturated colour, light bleeding through fog, figures half-consumed by shadow, carried through into the final room's wood-plank floor and material palette.

Literature

- T. Akenine-Möller, E. Haines, N. Hoffman, *Real-Time Rendering*, 4th ed., Ch. 9: N·L diffuse shading.
- I. Quilez, "Rendering: Soft shadows (raymarching)," iquilezles.org, 2010: the soft-shadow estimator actually used.
- J. C. Hart, "Sphere tracing," *The Visual Computer*, 1996: the sphere-tracing principle behind the raymarch.
- E. Reinhard, M. Stark, P. Shirley, J. Ferwerda, "Photographic tone reproduction for digital images," *ACM ToG*, 2002: the tone-mapping operator used.
- A. M. Turing, "Computing machinery and intelligence," *Mind*, 1950, and S. Turkle, *Alone Together*, 2011: framing for the ethical reflection on the antagonist's dual identity as AI assistant and rendering artifact.
- R. Arnheim, *Art and Visual Perception*, 1954: perceptual framing for the visual design.
- M. Grimshaw, "The audio uncanny valley," 2009: framing for the procedural audio design.
- M. M. Bradley, P. J. Lang, "Measuring emotion: the Self-Assessment Manikin," 1994: the evaluation instrument.

GPU Gems Vol. 1, Ch. 13 and Heikkinen and Holopainen were both in the v1 reading list and are deliberately absent here: neither ended up grounding anything actually implemented (the shadow technique used is entirely Quilez's, not GPU Gems' percentage-closer filtering), and Heikkinen and Holopainen's bibliographic record could never be independently verified as a stable, citable publication. Both were formally removed from the report's references with the reasoning documented in full there.

Current Status

- Rendering pipeline, game loop, and chase mechanic: complete and published.
- User study: complete, n = 8, moderated (in-person and synchronous screen-share), results written up in the report.
- Academic report: complete.
- Blog: 15 entries, published.